

Table of Contents	
1	Title
2	Block Diagram
3	MCXW7X MCU, RF, CLK
4	PWR MANAGEMENT
5	HMI, Sensors & Memory
6	I/O Headers - Ard, Mikroe, CAN, PMOD
7	MCU-Link
8	MCU-Link SWD

Revisions & Change Log			
Rev	Description	Date	Approved
X1	ISP drive high-state CAN support Ext Flash memory CS updated PMOD additon (DNP)	03/22/24	M.B.
A	Prototype Release	04/02/24	M.B.
A1	Matching network update - L3,L4,C50,C51 Antenna components update - R37, C54 Block diagram updated U1 MPN update	05/29/24 05/30/24 06/04/24	M.B.
B	J5 & J6 populated Production Release	06/18/24 07/12/24	M.B.

FRDM MCXW71



BL Secure Connected Edge
Connectivity
6501 William Cannon Drive West
Austin, TX 78755-8590

This document contains information proprietary to NXP and shall not be used for engineering design, procurement or manufacture in whole or in part without the express written permission of NXP Semiconductors.

© NXP SEMICONDUCTORS Classification: <FCP> <FILO> X

Designer:
ANTONIO QUIROZ

Drawing Title:
FRDM-MCXW71

Drawn by:
ANTONIO QUIROZ

Page Title:
TITLE PAGE

Approved:
MATTHIEU BAUDRY

Size
C

Document Number
SCH-93847 PDF: SPF-93847

Rev
B

Date:
Friday, July 12, 2024

Sheet
1 of 8

NOTES

1. Unless Otherwise Specified:
All resistors are in ohms, 5%, 1/8 Watt
All capacitors are in uF, 20%, 50V
All voltages are DC
All polarized capacitors are aluminum electrolytic
2. Interrupted lines coded with the same letter or letter combinations are electrically connected.
3. Device type number is for reference only. The number varies with the manufacturer.
4. Special signal usage:
_B Denotes - Active-Low Signal
<> or [] Denotes - Vectored Signals
5. Interpret diagram in accordance with American National Standards Institute specifications, current revision, with the exception of logic block symbology.

NOTES

This Color is for User Notes

This Color is for CAD/LAY Notes

This Color is for Silk

This Color is for DNP components

Target MCU: MCXW7x p[3]

Debug MCULink: LPC55S69JJEV98 p[7]

Short Table

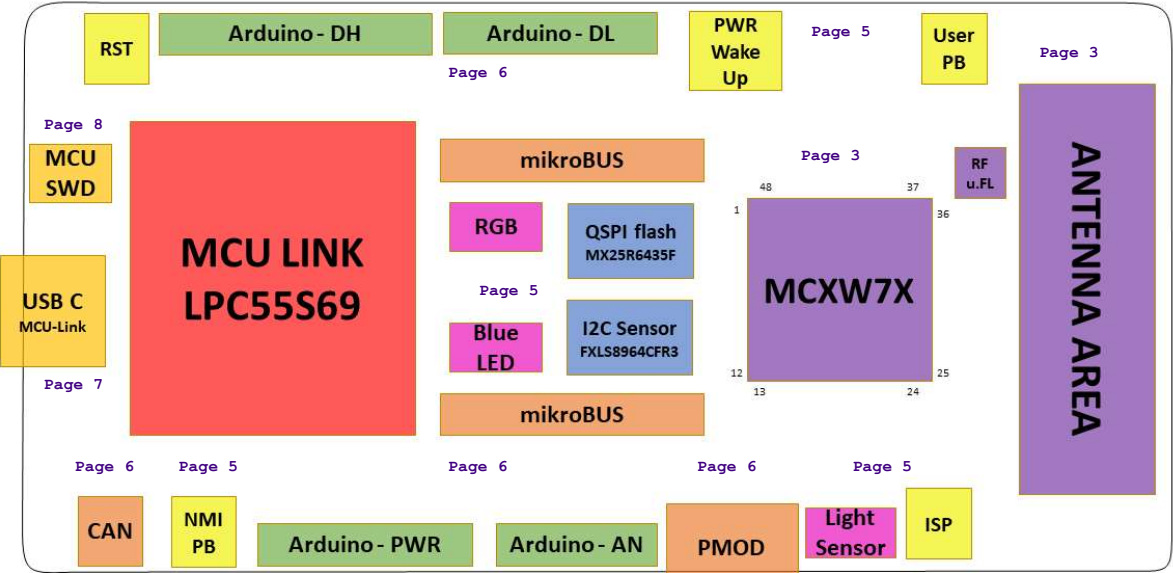
REF DES	SHORT	PAGE NAME
SH100,SH104,SH105,SH107,SH108,SH109,SH110,SH111,SH200,SH201	SHORTED	04. PWR MANAGEMENT
SH101,SH102,SH202,SH203	SHORTED	05. HMI, SENSOR & MEMORY
SH1,SH2,SH3,SH4,SH5,SH6,SH7,SH8,SH9,SH10,SH11,SH12,SH13,SH14,SH15,SH16,SH19,SH20,SH21,SH22,SH23,SH24,SH25,SH26,SH31,SH32,SH33,SH34,SH37,SH38,SH39,SH40,SH43,SH44,SH204,SH205,SH206,SH207,SH208,SH209,SH210,SH211	SHORTED	06. I/O Headers

DNP Table

REF DES	ASSY_OPT	PAGE NAME
C16,C17,C36,C37,C42,C53,C120,C121,C122,C123,TP9	DNP	03. MCXW7X
C20,C22,C30,D4,J11,J15,J16,J17,J20,JP1,JP2,R36,R64	DNP	04. PWR MANAGEMENT
C87,C100,C114,R4,R6,R8,R19,R162,R173,R174,R175,R187,R216	DNP	05. HMI, SENSOR & MEMORY
C110,C111,C112,C113,C127,J5,J6,J19,J22,JS1,JS2,JS3,JS4,R20,R28,R29,R97	DNP	06. I/O Headers
C78,C79,R130,R133,R137,U9	DNP	07. MCU Link
D9,D10,D11,R32,R33,R143,R153,R154,R155,R156,R157,R158,R159,R160	DNP	08. MCU Link SWD

Jumper Table

REF DES	JUMPER(DAFAULT)	JUMPER(DEFAULT)	PAGE NAME
JP4	1-2	1-2	04. PWR MANAGEMENT
JP1,JP2	NC	NC	04. PWR MANAGEMENT
JP10	1-2	1-2	05. HMI, SENSOR & MEMORY
JP9	1-2	1-2	07. MCU Link
JP5,JP6,JP7	OPEN	OPEN	07. MCU Link
JP8	1-2	1-2	08. MCU Link SWD



ICAP Classification: CP: IUX: X PUB:

Drawing Title:
FRDM-MCXW71

Page Title:
BLOCK DIAGRAM

Size C

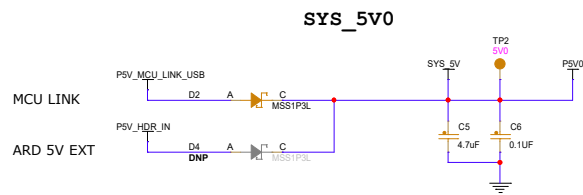
Document Number
SCH-93847 PDF: SPF-93847

Rev B

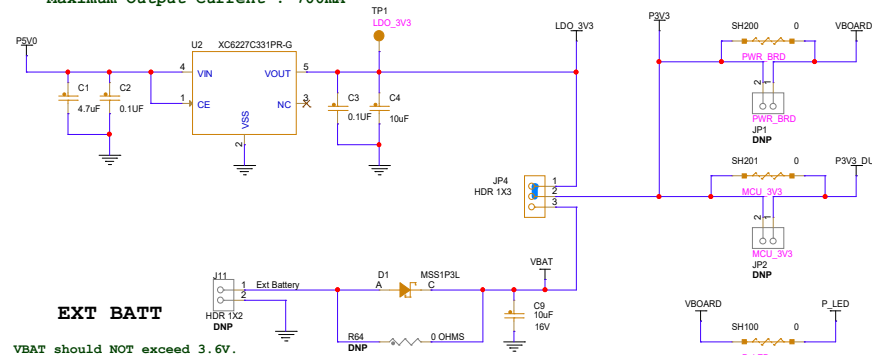
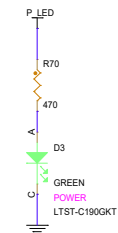
Date: Friday, July 12, 2024

Sheet 2 of 8

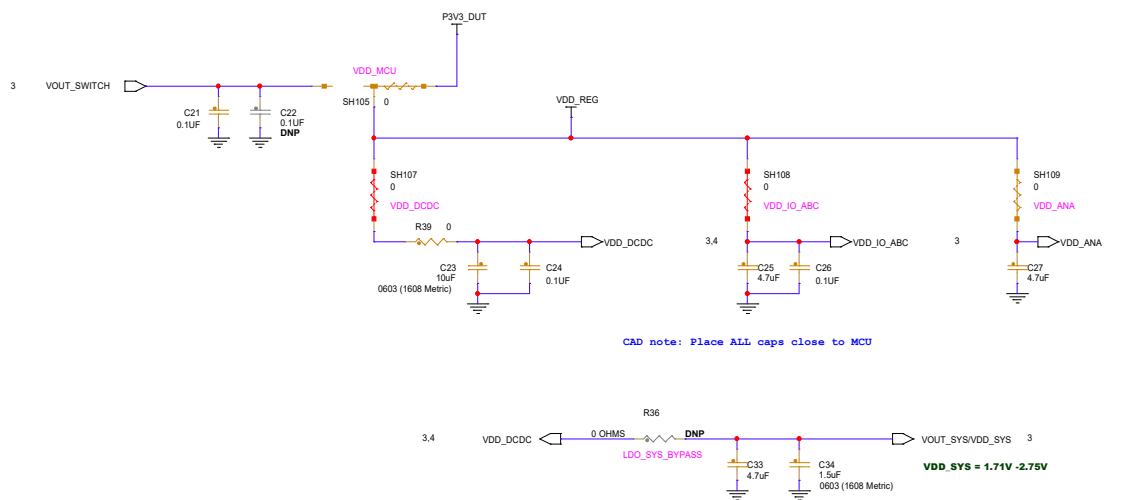
POWER MANAGEMENT



Maximum Output Current : 700mA

POWER ON
INDICATOR

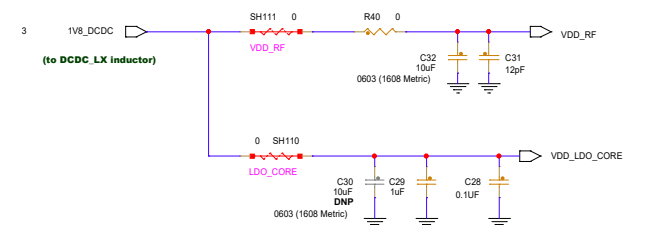
MCXW7X POWER CONFIGURATION



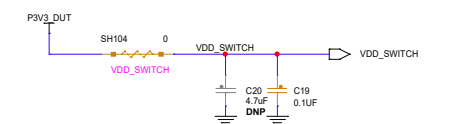
MCXW7X PWR CONFIG : DCDC Mode

VDD_SWITCH (SH104) : Shorted
VDD_DCDC (SH107) : Shorted
VDD_IO_ABC (SH108) : Shorted
VDD_ANA (SH109) : Shorted

```
1V8_DCDC : Shorted to LDO_CORE and VDD_RF
VDD_LDO_CORE (SH110) : Shorted
VDD_RF (SH111) : Shorted
```



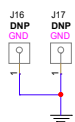
CAD note: Place ALL caps close to MCU



CAD note: Place ALL caps close to MCU

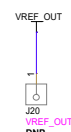
SH107, SH108, SH110 & SH111 used for current measurements

GND Test Points



Note:
Two ground test points are placed
on the left and right sides of the board to
facilitate testing

Power Supply Test Points



Mounting Holes

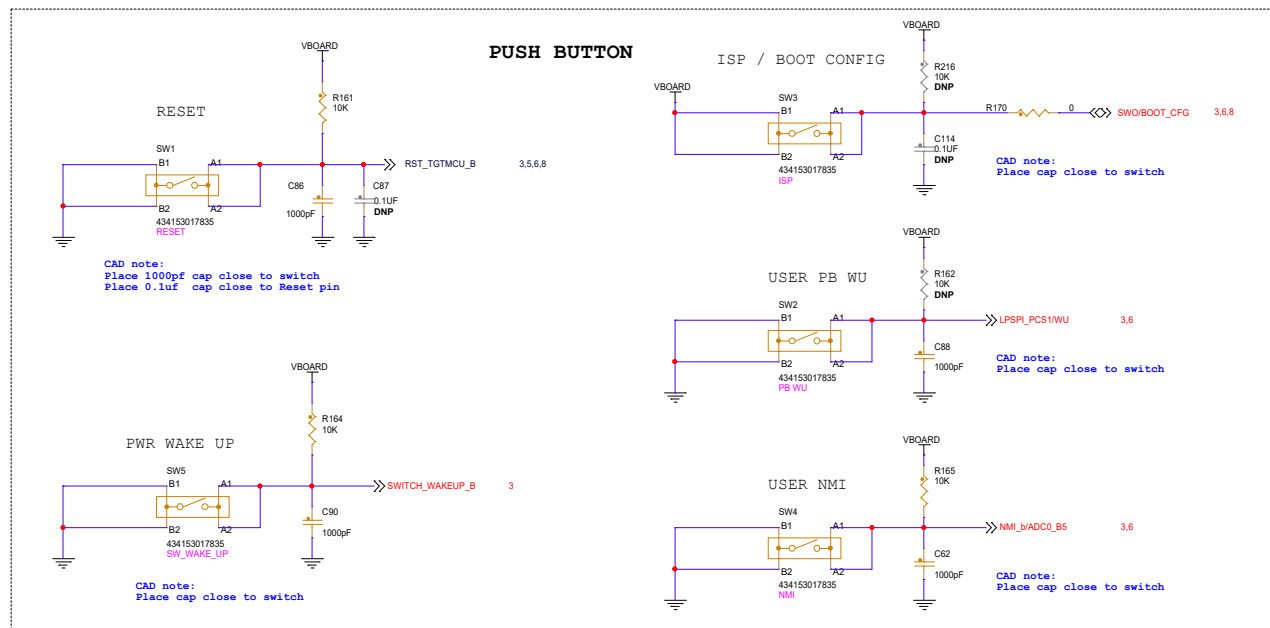


Note:
The mechanical mounting holes are
placed on the four vertices of the board



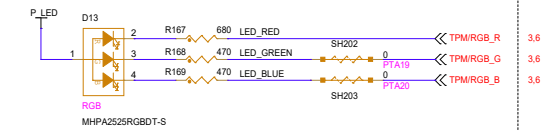
ICAP Classification: CP: _____ IUC: X PUBI: _____	
Drawing Title: FRDM-MCXW71	
Page Title: PWR MANAG	
Size C	Document Number SCH-93847 PDF: SPF-93847
Date: Friday, July 12, 2025	Sheet 4 of 8
Re B	

HMI



LEDS

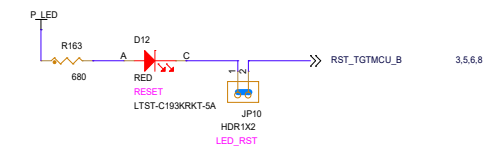
RGB LED



USER LED



RESET LED



64Mbit (8MB) FLASH

3.6,8 LPSP1_SOUT

3.6,8 LPSP1_SIN

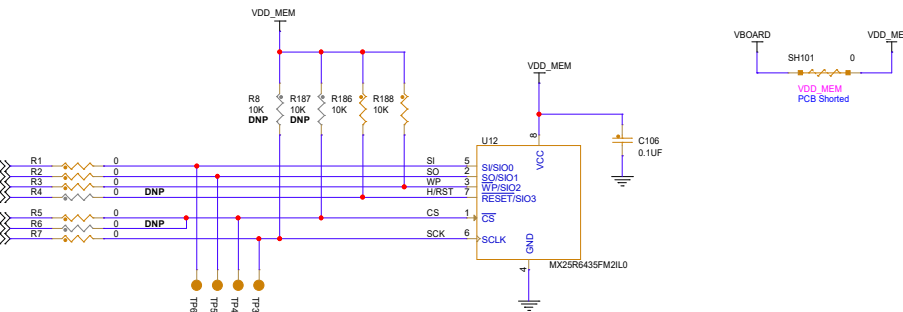
3.5,6 LPSP1_PCS3INT_SEN

3.5,6 LPSP1_PCS3LED_B

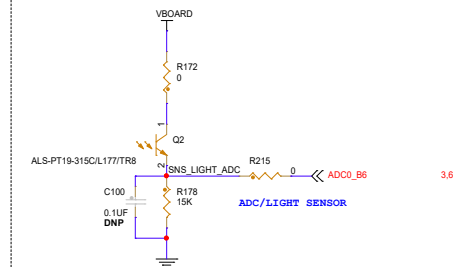
3.6,8 LPSP1_PCS0

3.5,6 LPSP1_PCS3LED_B

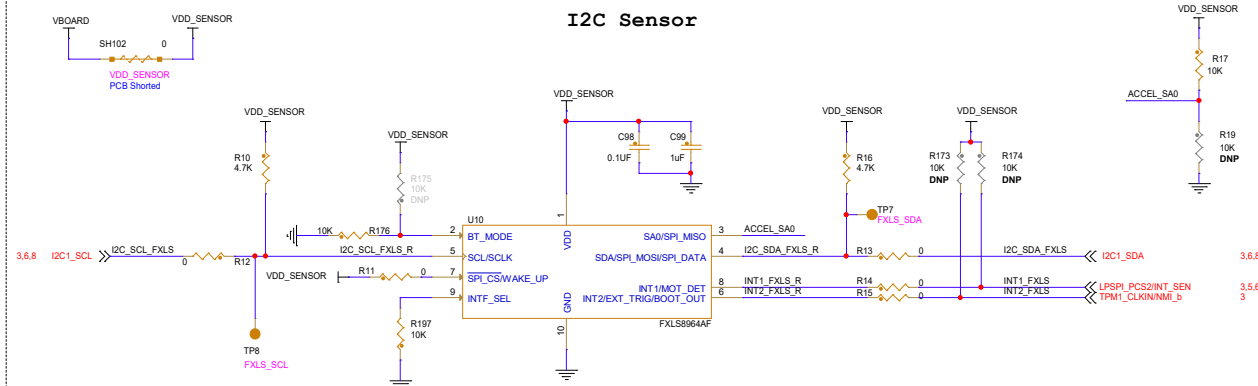
3.6,8 LPSP1_SCK



Ambient Light Sensor

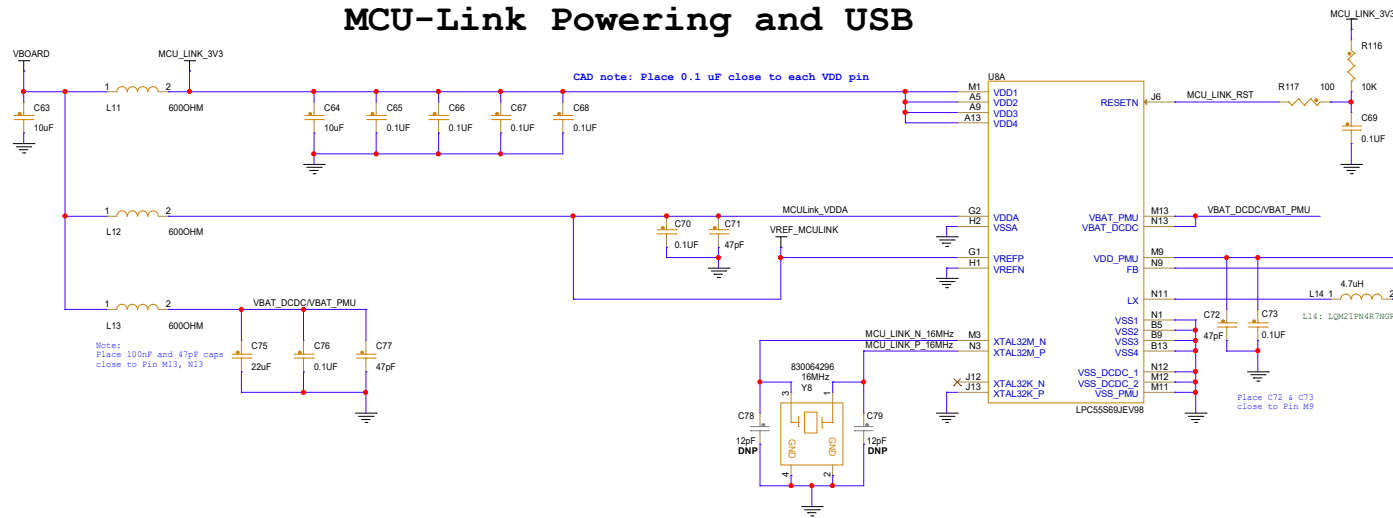


I2C Sensor

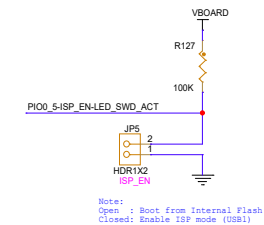
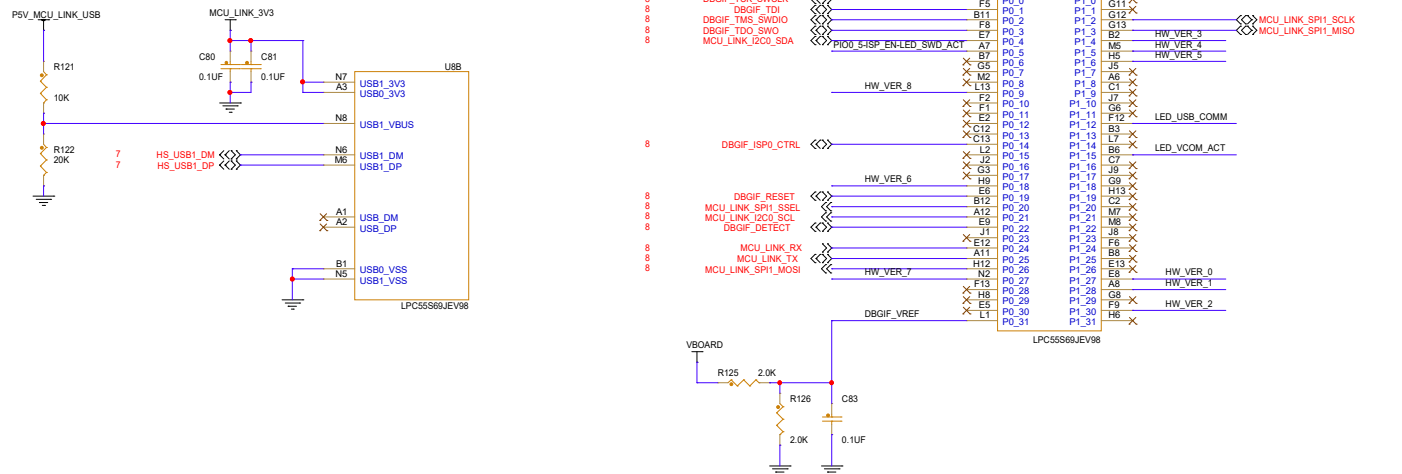
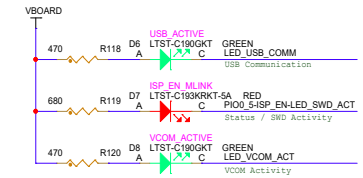


ICAP Classification: CP		IUC: X	PUB: B
Drawing Title: FRDM-MCXW71			
Page Title: HMI & MEMORY			
Size C	Document Number	SCH-03847 PDF: SPF-03847	
Date: Friday, July 12, 2024	Sheet	5	of 8

MCU-Link Powering and USB



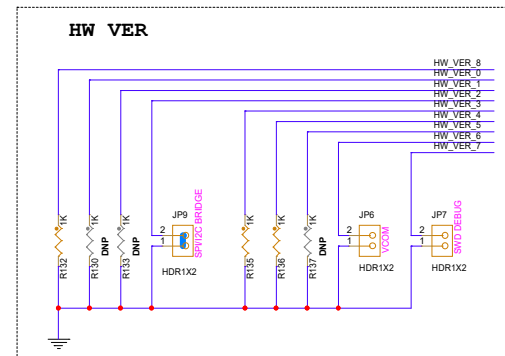
MCULINK LEDs



HW VER

HW VER	0	1	2	3	4	5	6	7	8
HW VER 0	PWR negotiation Low / Open								
HW VER 1	Reserved								
HW VER 2	SP1/I2C bridge	Open / Low							
HW VER 3	MCULink Version	Low / High							
HW VER 4	Board Identity	Low							
HW VER 5	PWR measurement	Low / Open							
HW VER 6	Primary VCOM	Open / Low							
HW VER 7	SWD Debug	Open / Low							
HW VER 8	Secondary VCOM	Open / Low							

HW VER



ICAP Classification:	CP	IUC:	X	PUB:	
Drawing Title:	FRDM-MCXW71				
Page Title:	MCU LINK				
Size C	Document Number	SCH-93847 PDF: SPF-93847			Rev B
Date:	Friday, July 12, 2024	Sheet	7	of	8

MCU LINK

